

NRES 746 – Week 4, Lecture 2: Expectation and the Sample Mean

Name: _____

Instructions: Work alone for about 8-10 minutes. Then compare with a neighbor.

Part 1: A parameter vs. a statistic

Suppose the length of brook trout in a stream (X) follows some distribution with pdf $f(x)$ and with true mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$ – these describe the *entire population* of trout in the stream, whether or not anyone ever catches one.

You catch $n = 5$ trout, measure them, and compute their average length $\bar{x}$.

1. Is μ a random variable? Is $\bar{x}$ a random variable? Briefly explain your reasoning.

Part 2: working with expectations

Here are some useful facts, true for any random variable X and constants a, b :

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \quad \text{Var}(aX + b) = a^2\text{Var}(X)$$

(notice that the shift b moves the mean but drops out of the variance entirely – shifting a distribution changes its location, not its spread; formally, this is because the variance of any constant is zero)

Also, for any two random variables X and Y :

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Suppose trout length X has $\mathbb{E}[X] = \mu = 22$ cm and $\text{Var}(X) = \sigma^2 = 16$ cm².

2a. What is $\mathbb{E}[2X + 3]$? What is $\text{Var}(2X + 3)$?

$\mathbb{E}[2X + 3] =$ _____ $\text{Var}(2X + 3) =$ _____

2b. A second, independent measurement Y of a different trout has the same mean and variance as X . It is always true that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$. It is also true in this case that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. However, this second fact is not always true. What do you think might be the special condition that makes it true?

Part 3: Averaging two measurements

Let X_1 and X_2 be two independent measurements of the same quantity, each with mean μ and variance σ^2 . Define their average $M = \frac{X_1 + X_2}{2}$.

3a. Rewrite M as $aX_1 + bX_2$ (i.e. what are a and b ?), then use the facts from Part 2 to derive $\mathbb{E}[M]$ and $\text{Var}(M)$ in terms of μ and σ^2 . Show every step.

3b. Using $\mu = 22$ and $\sigma^2 = 16$ (from Part 2), what is $\text{Var}(M)$ as a number? Compare it to $\text{Var}(X_1) = 16$ alone – how does the variance of an average of two measurements compare to the variance of each of its constituent parts?

$\text{Var}(M) =$ _____

Part 4: Generalizing to n measurements

Now generalize: let $X_1, X_2, \dots, X_n$ be independent measurements, each with mean μ and variance σ^2 , and define the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

4a. Following the same logic as Part 3 (now with n terms instead of 2), derive $\mathbb{E}[\bar{X}]$ and $\text{Var}(\bar{X})$ in terms of μ , σ^2 , and n .

4b. Check your formula against $n = 5$, $\sigma^2 = 16$: what number does $\text{Var}(\bar{X})$ come out to? How does it compare to Part 3b when you plug in $n = 2$?

$\text{Var}(\bar{X}) =$ _____

4c. As $n \rightarrow \infty$, what happens to $\text{Var}(\bar{X})$? What does that mean, in plain language, about how precisely you can pin down μ with a very large sample?

4d. One more thing to be careful about: $\text{Var}(\bar{X})$ and the *standard deviation* of $\bar{X}$ (often called the standard error) are not the same formula. If $\text{Var}(\bar{X}) = \sigma^2/n$, write down the standard deviation of $\bar{X}$ – how does it scale with n ?
